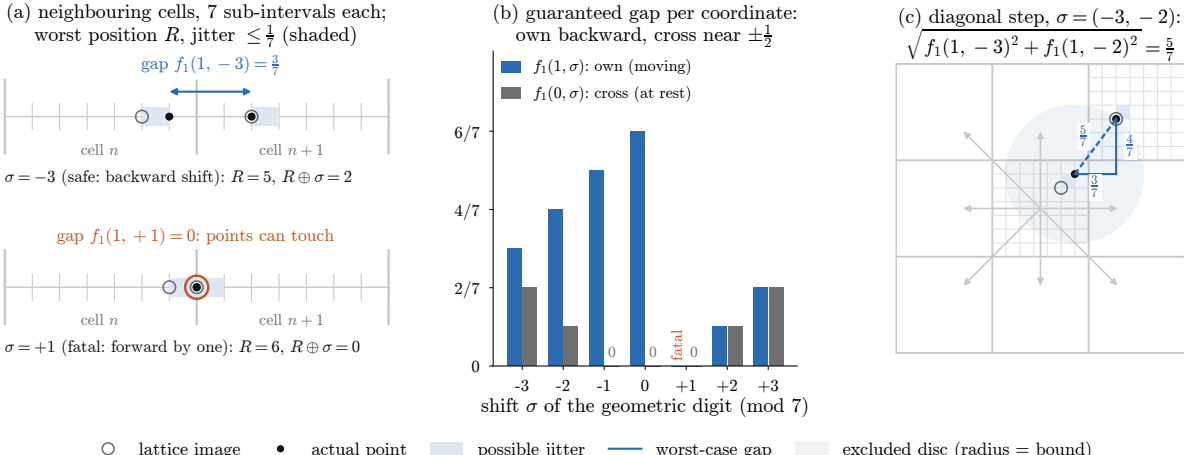
(a) neighbouring cells, 7 sub-intervals each; worst position R, jitter $\leq \frac{1}{7}$ (shaded)
gap $f_1(1,-3)=\frac{5}{7}$
cell n
cell $n+1$
$\sigma=-3$ (safe: backward shift): $R=5$, $R\oplus\sigma=2$
gap $f_1(1,+1)=0$: points can touch
cell n
cell $n+1$
$\sigma=+1$ (fatal: forward by one): $R=6$, $R\oplus\sigma=0$
(b) guaranteed gap per coordinate: own backward, cross near $\pm\frac{1}{2}$
$f_1(1,\sigma)$: own (moving)
$f_1(0,\sigma)$: cross (at rest)
6/7
4/7
2/7
0
0
0
0
fatal
-3
-2
-1
0
+1
+2
+3
shift σ of the geometric digit (mod 7)
(c) diagonal step, $\sigma=(-3,-2)$: $\sqrt{f_1(1,-3)^2+f_1(1,-2)^2}=\frac{5}{7}$
lattice image
actual point
possible jitter
worst-case gap
excluded disc (radius = bound)